UNIT 1
INTRODUCTION TO STATISTICS

ORIGIN

- The word statistics as been derived from:

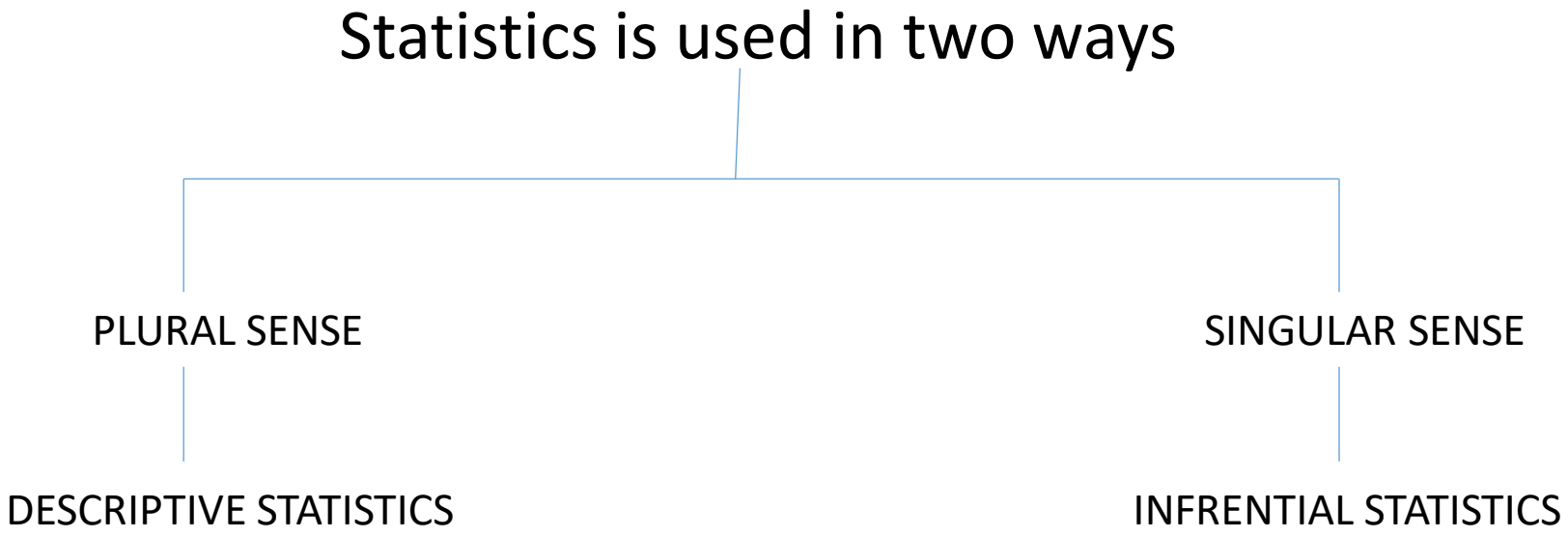
The Latin Word : “status”

The Italian Word : “statista”

The German Word : “statistic”

Each of these words means a “political state”.

MEANING


Statistics is used in two ways
PLURAL SENSE
SINGULAR SENSE
DESCRIPTIVE STATISTICS
INFRENTIAL STATISTICS

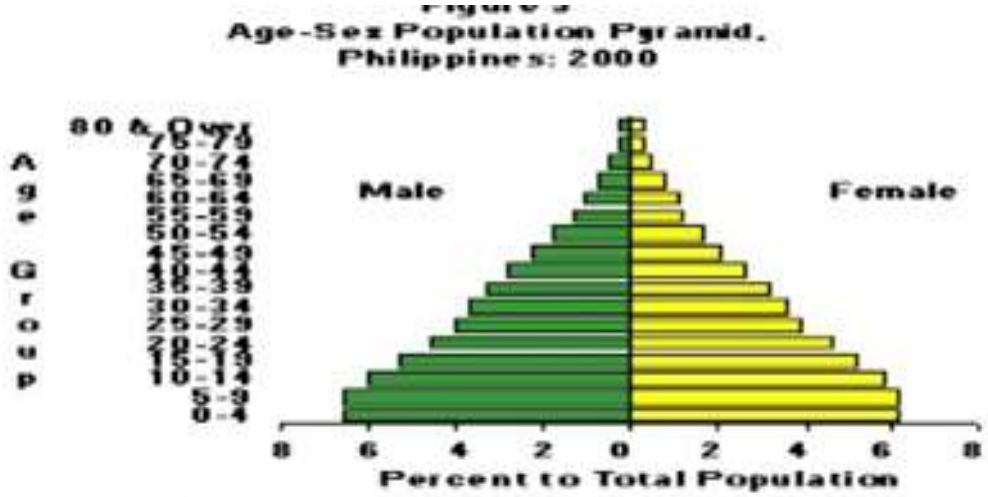
DESCRIPTIVE V/S INFERENTIAL

- DESCRIPTIVE STATISTICS : It includes the procedure for collecting , classifying, summarizing and presenting data.

  It enables us to evaluate critically the information presented in reports, articles , etc.
- INFERENTIAL STATISTICS : It not only describes the important features of the data but also allows to proceed beyond data collection into the area of decision making through generalizations and predictions.

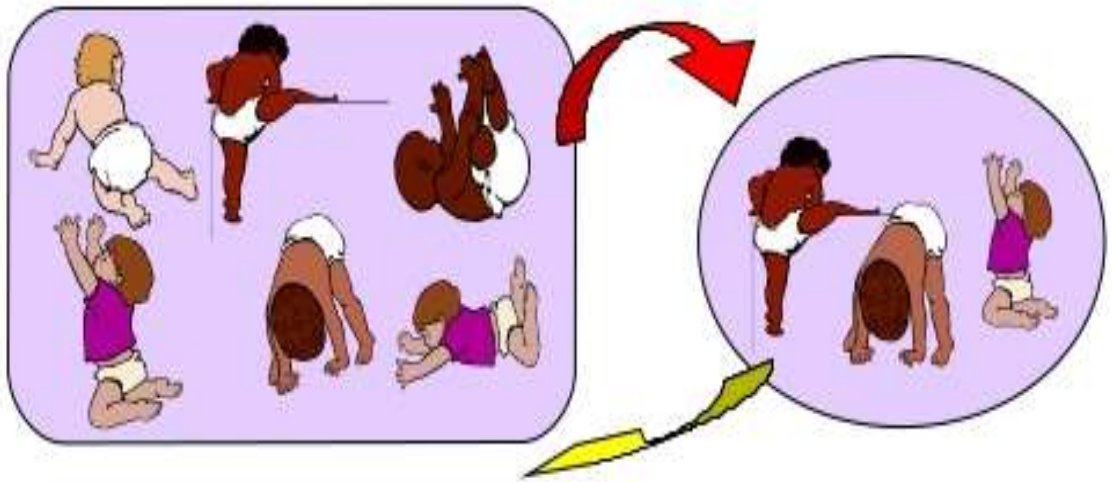
EXAMPLE OF DESCRIPTIVE STATISTICS

The Philippine population by constructing a graph indicating the total number of Filipinos counted during the last census by age group and gender



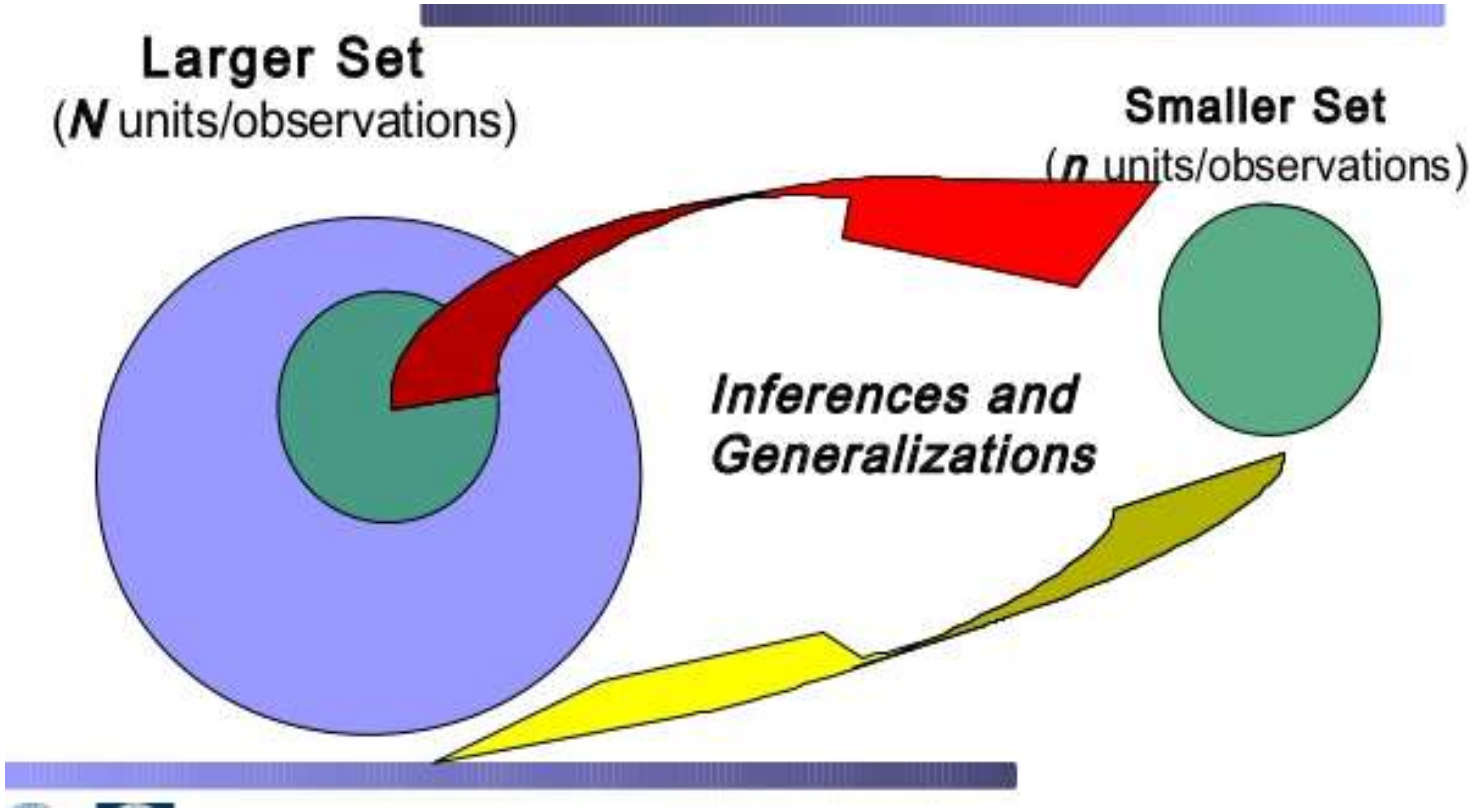
EXAMPLE OF INFERENTIAL STATISTICS

A new milk formulation designed to improve the psychomotor development of infants was tested on randomly selected infants.



Based on the results, it was concluded that the new milk formulation is effective in improving the psychomotor development of infants.

INFERENTIAL STATISTICS



DEFINITIONS

- plural sense: numerical facts, e.g. CPI, rupee-dollar exchange rate
- singular sense: scientific discipline consisting of theory and methods for processing numerical information that one can use when making decisions in the face of uncertainty.

DEFINITIONS

- Statistics may be defined as the science of collection, presentation, analysis and interpretation of numerical data.

  - Croxton and Cowden

- Statistics is the science of estimates and probabilities.

  - Boddington

- Statistics may be called the science of counting and averages.

  - Bowley

CHARACTERISTICS

- **Statistics are aggregate of Facts** : A single fact or figure cannot be called statistics.
- **Statistics are expressed numerically** : Statistics is the study of those facts which can be expressed numerically.
- **Statistics must be placed in relation to each other**: The statistical data which is collected must be comparable with each other.
- **Statistics are collected for a pre- determined purpose** : The purpose and objective of collecting the data must be clearly defined.

NATURE

IS STATISTICS A SCIENCE OR AN ART OR BOTH ?

- **Science** refers to a systematized body of knowledge, it studies causes and effect relationship to find out generalizations.
- **Cowden** defined, “statistics as the science of collection, presentation, analysis and interpretation of numerical data”.
- **Art** is that branch of knowledge which tells us the best possible methods so as to meet certain specifications.
- According to **Harlow**, “Statistics is both a science and an art of handling aggregate of facts-observing, enumeration, recording, classifying and otherwise systematically treating them.

FUNCTIONS

- **PRECISENESS AND DEFINITENESS**: Statistical techniques enable us to present facts in clear, precise and definite form.
- **CONDENSATION**: With the help of statistical tools a mass of data can be condensed into a few presentable, understandable and significant figures.
- **COMPARISON**: the absolute figure themselves do not convey any significant meaning. It is their comparison tat helps us to draw conclusions.
- **FORMULATION AND TESTING OF HYPOTHESIS**: Establishes a relationship between two or more variables.
- **PREDICTION:** Analyse past data and predict future trends
- **FORMULATION OF SUITABLE POLICIES:** helps in formulation of various economic, business and other national policies.

USE OF STATISTICS

Statistics is one of the most powerful and indispensable tools in any specific enquiry which has found applications to a very large number of disciplines, such as:

- Agriculture
- Industry
- Psychology
- Ecology
- Economics
- Insurance
- Business
- Sociology
- Biology

Statistics in Business, Commerce and Industry

- Decision Making
- Time Series Analysis
- Analyze demand
- Business Cycle Trend

Statistics in Economics

- Economic Problems: unemployment, poverty, prices, income distribution
- Statistical data and techniques
- Economic Theories

Statistics in Science

- Astronomy
- Agriculture
- Meteorology
- Biology
- Medicine

Statistics in Research

- Development of New Techniques
- Strategies
- Theories

LIMITATIONS

- **Qualitative Aspect is Ignored**: It needs conversion of qualitative data into quantitative data. Eg : Health, Stress, Beauty
- **It does not deal with individual items**: statistics deals with only aggregates of facts or items and it does not recognize any individual item.

  Eg: Expenditure incurred by a family in a month, average marks of BCA III in statistics.

- **It does not depict entire story of phenomenon :** When ever a phenomena happen, that is due to many causes, but all these causes can not be expressed in terms of data. So we cannot reach at the correct conclusions.

  Eg: Demand of Commodity

- **It is liable to be miscued :** As W.I. King points out, “One of the short-comings of statistics is that do not bear on their face the label of their quality.”

  Eg: Error in collecting data (bias)

- **Results are true only on average:** “Statistics largely deals with averages and these averages may be made up of individual items radically different from each other.” —W.L King
- **Too Many methods to study problems:**

  In this subject we use so many methods to find a single result, which may cause variations in the results.

- **Statistical Interpretations require a high degree of skill and understanding of the subject.**

Thank You..!